

SHALON FERNANDO

SOFTWARE ENGINEER INTERN

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Software Engineer with a rare ability to bridge modern UI architecture (**Next.js/React**) with complex **ML and enterprise backends (Python, Spring Boot)**. Proven track record of shipping **production-ready B2B SaaS platforms** and securing **top-3 finishes in national hackathons**. Currently engineering full-stack architecture ecosystems as an intern at **ClusterX**, while independently architecting **low-latency fraud detection and FinTech applications**.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, SQL, C#

Backend: Spring Boot, Node.js, Express.js, Flask, Spring Security, RESTful APIs, .NET 9, ASP.NET Core, MCP

Frontend: React, Next.js (App Router), React Native (Expo), Tailwind CSS, Three.js

AI/ML: XGBoost, SHAP, Hugging Face Transformers, LangChain, RAG, OpenAI Whisper, Gemini AI, FAISS, Multi-Agent Orchestration, GenAI SDK, AI Agents

Databases: MySQL, MongoDB, PostgreSQL, Firebase Firestore, Supabase

Cloud & DevOps: Docker, GitHub Actions(CI/CD), Vercel, Render, Kubernetes, Microservices Architecture, n8n

Testing & Tools: JUnit 5, Mockito, JaCoCo, Postman, Agile/Scrum, Jira, Git

PROFESSIONAL EXPERIENCE

Software Engineer Intern - [ClusterX LLC](#) | Remote | Dec 2025 to Present

- Dentiflo:** Contributed full-stack updates to a dental inventory platform using Next.js/React, optimizing UI architecture, onboarding, and SEO performance.
- Accflow:** Architected key accounting workflows for UK clients, including client onboarding, AML compliance, document tracking, and billing automation.
- Full-Stack & APIs:** Integrated enterprise third-party services including Xero, QuickBooks, Microsoft API, and secure user flows via Better Auth.
- Code Quality:** Authored scalable TypeScript components, managed frontend logic patterns, and collaborated on code reviews within strict enterprise standards.

Fullstack Software Engineer – Intern Zenvora Technologies Pvt. Ltd. | Remote | Oct 2025 - Dec 2025

- Tasked with architecting the core AI/ML summarization engine for an automated document filtering system for a major banking client (DFCC Bank).
- Conducted a comprehensive case study on AI-driven mandate processing; presented the full technical architecture and NLP pipeline strategy (RAG/Transformers) directly to the CTO.

Software & Digital Transformation Engineer(Part-time) - Cookie Love | Nov 2025 – Present

- Building a staff-focused order-management mobile app to prevent missed orders and improve kitchen workflow.
- Developing the Cookie Love website to support digital menus and upcoming online ordering features.
- Digitizing daily order tracking and inventory records to replace paper-based processes.

Technical Writer (Contributor) | [gitconnected](#) (Level Up Coding) | April 2026 – Present

- Selected as a recurring author for a top-tier technical publication with 315k+ global followers.
- Architecting deep-dive articles on MLOps, high-performance FastAPI backends, and FinTech security.

PUBLICATIONS

Technical Author | [Level Up Coding](#) ([gitconnected](#)) - April 2026

- Published an architectural deep-dive titled "[I Built a Fraud Detection System That Explains Itself](#)," reaching a global community of 315k+ developers. Articulated the engineering behind a sub-85ms inference engine using FastAPI and XGBoost. Documented the implementation of SHAP (explainable AI) to make high-stakes ML decisions transparent and human readable.
- Published an security-focused engineering deep-dive titled "[Locking Down Your App The Better Auth Way](#)." Evaluated modern 2FA (TOTP/OTP) protocols, multi-provider account linking strategies, and granular session control mechanics to eliminate high-risk vulnerabilities across enterprise application architectures.

PROJECTS

FraudGuard AI - Enterprise Fraud Risk Engine | [Live Demo](#) | [GitHub](#)

Architected a **B2B SaaS platform** that scores **financial transactions for fraud in under 85ms** using a custom **XGBoost** inference engine. Built a high-performance Next.js dashboard and a **Python/FastAPI** backend, utilizing FastAPI lifespan events for zero-latency ML model booting. Implemented white-box ML via **SHAP** to generate granular risk factor explanations, alongside a hybrid rules engine, **Supabase** telemetry logging, and **Stripe** billing integration.

ResQ Hub – Sri Lanka Emergency Response Platform | [Live Demo](#) | [GitHub](#)

(1st Runner-up, Codefest 2025) Built a real-time emergency response platform for the 2025 Sri Lanka floods, integrating **SOS alerts, RouteWatch, relief requests, missing-person search, flood analytics, and an AI Disaster Assistant**. Powered by national-level data APIs (**DMC Flood API, geolocation, and real-time sockets**). Designed responder and admin dashboards for incident verification, resource allocation, and live communication.

CareLink – AI-Powered Cloud-Native Healthcare Platform | [Live Demo](#) | [GitHub](#)

Engineered a cloud-native microservices architecture using **.NET 9**, **PostgreSQL**, and **YARP API Gateway** for secure request routing. Developed an **AI Symptom Checker using XGBoost** and an **AI Chatbot** to streamline doctor discovery and patient management. Built a real-time **Telemedicine module** for video consultations, alongside secure payment processing and notification services. Containerized and deployed the full ecosystem via **Docker and Kubernetes (K8s)** for scalable production environments.

Ruki – Voice Activated Multi-Agent AI E-Commerce Copilot for Kapruka | [Live Demo](#) | [GitHub](#)

Architected a voice-first e-commerce assistant for Kapruka using Next.js, FastAPI, and PostgreSQL. Engineered a multi-agent routing system powered by Vertex AI to orchestrate specialized UI and catalog agents. Implemented localized NLP supporting cross-lingual queries (Tanglish/Sinhala) and a multimodal visual search pipeline. Developed custom agentic workflows for group gifting links, item comparisons, and live checkout sync.

LegalAid AI - AI Powered Legal Assistant | [Live Demo](#) | [GitHub](#)

Architecting a multimodal legal assistant using a **Next.js** (TypeScript) frontend and a **FastAPI** (Python) backend. Designed to answer legal queries via a web UI and WhatsApp, integrating an **OpenAI Whisper model** for voice-to-text transcription. Implements a RAG pipeline (LangChain, FAISS) for document retrieval and an **n8n workflow** automation engine for scheduling tasks.

MediWay – Hospital Management System | [Live Demo](#) | [GitHub](#)

Developed a full-stack hospital platform with a **Spring Boot** backend, achieving **82% test coverage** across 484 unit and integration tests (JUnit 5, Mockito, JaCoCo). Engineered a scalable enterprise architecture, integrating Docker, PayPal API for payments, and **GitHub Actions (CI/CD)** for a complete development-to-deployment workflow.

HACKATHONS & LEADERSHIP

- **Winner – MS Club Mini Hackathon** - Secured 1st place after advancing through the engineering prototype round, delivering an innovative and well-executed solution in the final stage. - [Certificate](#)
- **Team Lead – 1st Runner Up, ReviveNation Hackathon** (Codefest 2025, SLIIT) – Led a team to design and develop an innovative AI-driven disaster management solution under tight deadlines - [Certificate](#)
- **2nd Runner Up | IntelliCon'25 hackathon | AIESEC in SLIIT** - Awarded 2nd Runner Up for developing a legal aid AI application that delivered an innovative solution to a complex problem. - [Certificate](#)
- **Finalist, VibeFlow AI Hackathon** by STEM LINK – prototyped edtech web app with documentation in 6h - [Certificate](#)
- **Finalist, HackX 10.0 | IdeaX 2025** – Competed in the final stage after initial selection among top teams for innovation pitch. - [Certificate](#)

EDUCATION

B.Sc. (Hons) in Information Technology - Specializing in Software Engineering Sri Lanka Institute of Information Technology (SLIIT) | (Expected Graduation)

- **Dean's List Honoree** (Year 1, Semester 2) - [Certificate](#)

CERTIFICATIONS

1. **Getting Started with AWS Cloud Essentials** — AWS/Skillbuilder.
2. **Agile Scrum Foundation** — Simplilearn (Issued Jul 2025) Credential ID: 8557338
3. **Hugging Face Projects** — Simplilearn (Issued Jul 2025) Credential ID: 8647177
4. **JavaScript Algorithms and Data Structures** — freeCodeCamp (Issued May 2025) Credential ID: fcc-62c83883-49b6-4a77-ad3b-10e101db8211-jaads
5. **MongoDB Data Modeling Path** — MongoDB (Issued Oct 2025) Credential ID: MDBganoz9dqjy
6. **Postman API Fundamentals Student Expert** — Postman (Issued Nov 2025) ID: 690a01291243d3353edbf0e9
7. **AI Driven Full-Stack Bootcamp** — STEMLINK (Issued NOV 2025) ID: cef164d3-7727-4022-babf-a6f7b6f4f31c

[View all 29+ certifications on LinkedIn](#)

REFERENCES

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